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Pathfinder

Integrating GenAI with Graph Database for the Enterprise

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Analyst View on Gen AI, Graph DB & Ontologies

By 2025,

10% of all global data produced will come from generative AI.

Users expect answers to the questions, not list of links to resources that might get the answers

75% of Fortune 500 companies are actively engaged in semantics initiatives as part of the data strat

By 2025,

graph technologies will be used in 80% of data and analytics innovations

Generates human-like text based on input prompts or context.

Generates, Image, Music and Video as per the prompts provided

Generates synthetic data to augment existing datasets for training machine learning models.

Generates 3D models for designs, virtual environments, and simulations.

Generates human-like speech and can be used for text-to-speech synthesis, voice assistants,

Text Generation

Image, Music and Video Generation

Data Augmentation

3D Model Generation

Speech Generation

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Pathfinder - Use Case

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Gen AI – Challenges it poses

Implicit Knowledge / Knowledge Cut off

Hallucination

Indecisiveness

Lack of enterprise domain knowledge

Reasonable answers but not always accurate

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Shortcomings of Traditional RDBMS

SQL models tables and keys, not business concepts

SQL lacks support for complex types

SQL has poor support for abstraction and reuse

SQL schemas are rigid, its difficult to add new relationships without schema surgery

No good way to express business rules and constraints

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Why use Graph for Gen AI ?

Structured Data & Relationships

Explainability and Traceability

Support more diverse and complex queries

Consistency Vs Hallucination

Low Latency Retrieval

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What is a Knowledge Graph ?

A knowledge graph is a database, that models business concepts, the relationships between them and t

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Knowledge Graphs - Triples

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Gen AI for Knowledge Graph Creation

Use LLMs for creating of Ontologies which will act as a guide

Identify and resolve entity mentions in text by using NLP – Named Entity Recognition (NER)

Extract semantic relationships from text data, uncover implicit relationship between entities

By integrating Natural Language Understanding (NLU) models with KGs, interpret complex queries expre

Help to streamline the process of importing data into KG

Gen AI Models can analyze patterns and trends to make predictions and recommendations

On the Fly Ontologies■

Entity Resolution

Relationship Extraction

Query Understanding

Data Integration

Predictive Analysis

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Feature Enrichment of Data through GDS

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Feature Enrichment – Sample Algorithms

Page Rank

What : Finds important nodes based on their relationships

Why : Identify important or influential client nodes by quantifying the flows of money towards them.

Uses :

Fraud detection

Anti-money Laundering

Inform prioritization during analysis and investigation

Weakly Connected Components

What : Finds disconnected community subgraphs in our data

Why : Identify communities based on connections with shared pieces of identity

Uses :

Householding

Synthetic identities

Stolen identities

Node Similarity

What : Similarity between nodes based on neighbors. Writes a new relationship to the graph

Why : Identify similar nodes who share common pieces of identity

Uses :

Entity Resolution

Synthetic identities

Stolen identities

Louvain Modularity

What : Finds communities in our graph who are connected. Can return intermediate results

Why : Useful for identifying communities based on transaction behavior rather than identity

Uses :

Fraud ring detection

Anti-Money Laundering

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Feature Enrichment – Sample Variables

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Book of work done through Graph Database